Empirical Dual-Slope Propagation Modeling of 915 MHz LoRa in Hyper-Dense Urban Canyons: A Waveguide Characterization

Radoslaw Green, Divyam Joshi

***Abstract*—Sub-GHz Low Power Wide Area Networks (LPWANs) are increasingly deployed in dense urban environments, yet standard propagation models often fail to accurately predict their performance in high-rise corridors. This paper investigates a pronounced radio frequency propagation anomaly at 915 MHz within the hyper-dense "urban canyons" of downtown Singapore, where signals demonstrate unexpected survivability over extreme distances. Using a custom LoRa measurement platform based on the SX1262 transceiver, we empirically characterize this phenomenon, identifying a distinct waveguide-like regime ($n_1 \approx 1.4$) that deviates from standard single-slope cellular models ($n \approx 4.0$). Rejecting oversimplified aggregate path loss assumptions, we apply a dual-slope ITU-R P.1411 framework to calculate a pre-breakpoint waveguide exponent of $n_1 \approx 1.4$ and a post-breakpoint attenuation of $n_2 \approx 4.95$, helping to explain the empirical "Reverse S-Curve" drop-off. We further provide a stochastic derivation of outage probability to validate the physical limits of the link. As a corollary to these propagation findings, we evaluate the system-level implications for decentralized mesh networks using a validated ns-3.41 Monte Carlo simulation framework, characterizing flood-routing percolation thresholds across the 42-segment OSM Singapore CBD street graph. The waveguide model percolates at $N_c \approx 350$ nodes versus $N \approx 500$ for the standard $n = 4.0$ model, requiring 30% fewer relay nodes and achieving 2.5× fewer hops at peak density. These findings provide critical empirical correction factors for propagation simulators, bridging the gap between theoretical urban attenuation models and observed sub-GHz physical resilience.**

***Index Terms*—LoRa, Internet of Things (IoT), Urban Propagation, Spatial Characterization, Mesh Networks, Geometric Attenuation, Stochastic Modeling, Urban Canyon.**

I. Introduction

THE deployment of Sub-GHz Low Power Wide Area Networks (LPWANs), particularly Long Range (LoRa) technology, has fundamentally altered the paradigm of urban Internet of Things (IoT) connectivity. However, the physical layer (PHY) propagation characteristics of LoRa in hyper-dense, highly structured urban matrices remain insufficiently modeled. Standard telecommunications simulators routinely employ single-slope Log-Distance path loss models, typically assigning a Path Loss Exponent (PLE) of $n \approx 3.5$ to 5.0 for dense urban environments [22]. Under these mathematical assumptions, a 915 MHz signal should experience near total terminal attenuation within a 3 to 4-kilometer radius. Yet, empirical observations frequently contradict these limits, demonstrating successful packet delivery at distances exceeding 10 kilometers in heavily built-up city centers.

The authors are with the Canadian International School, Singapore. (e-mail: ragr2028@cis.edu.sg).

This study isolates and investigates this propagation anomaly. We hypothesize that the linear geometric arrangement of modern concrete and steel high-rises creates a waveguide-dominated propagation regime, channeling RF energy and drastically reducing the decay rate along specific arterial corridors [23]. To test this, we engineered a custom measurement platform based on the Broadcom BCM2710A1 processor and Semtech SX1262 architecture to rigorously map the geometric path loss profile of downtown Singapore.

Understanding the precise mechanics of this waveguide anomaly is not merely a theoretical exercise; it has immediate, critical implications for the design of decentralized, gateway-free mesh networks. In scenarios where centralized telecommunications infrastructure experiences large-scale outages—such as the 2011 Great East Japan Earthquake or the 2021 Western Kentucky tornadoes—emergency response relies entirely on ad-hoc mesh connectivity [1], [2], [36]. The viability of these routing algorithms is intrinsically bound to the electromagnetic realities of the substrate upon which they operate. If standard models overestimate attenuation, network engineers will massively over-deploy hardware; conversely, failing to account for sharp non-Line-of-Sight (NLoS) diffraction penalties will lead to severe network partitioning.

By providing high-fidelity empirical data and applying a dual-slope stochastic mathematical framework, this research seeks to explain the contradiction between standard path loss models and observed LoRa performance. Furthermore, we translate these physical layer findings into system-level constraints, evaluating how nodes must be architected in terms of power autonomy and spectral resilience to exploit this waveguide phenomenon in grid-denied environments.

II. Related Work

The transition from centralized IoT monitoring to edge-heavy, decentralized architectures requires a precise understanding of sub-GHz performance boundaries.

A. LPWAN Topologies and Simulation Artifacts

Technologies such as LoRaWAN and Narrowband IoT (NB-IoT) currently dominate the smart city landscape [3], [4], [34]. However, standard LoRaWAN deployments are anchored by a strict "star-of-stars" topology. As detailed by Haxhibeqiri et al. [5], edge devices broadcast blindly to centralized gateways. Adelantado et al. [6] demonstrated that the unslotted ALOHA-based MAC protocol utilized by

standard LoRaWAN restricts network capacity due to high collision probabilities, a fundamental scalability ceiling further characterised by analytical capacity studies [7] and large-scale ns-3 scalability analyses [8], [10] and comprehensive scalability surveys [9].

To circumvent these structural bottlenecks, the research community has pivoted toward "gateway-free" mesh architectures [11], [12]. Protocols like LoRaBlink [13] and ESP32-based mesh algorithms [14] have validated the computational feasibility of distributing routing intelligence to the edge, and empirical multi-sensor deployments have demonstrated the coverage advantages of such topologies over single-hop star networks [15]. However, the simulation of these routing protocols frequently relies on idealized path loss assumptions, creating a validation gap when deployed in complex urban substrates.

B. Empirical Urban Propagation and 2.4 GHz Limitations

While 2.4 GHz mesh protocols (Zigbee, Wi-Fi) are ubiquitous, electromagnetic physics dictates they suffer from severe free-space path loss and diffraction penalties, failing to penetrate concrete canyons [16], [17]. Conversely, existing literature on sub-GHz LoRa propagation has featured varied topographies. Petäjäjärvi et al. [18] conducted extensive 868 MHz LoRa measurements across the city of Oulu, Finland, deriving an empirical channel attenuation model with ground-level path loss consistent with a conventional sub-urban regime ($n \approx 2.1$, exceeding free-space), while Augustin et al. [19] measured PLEs of $n \approx 3.8$ in the built-up Paris region, reflecting stronger urban attenuation.

Singapore represents a unique RF challenge, possessing a building density that creates deep "Concrete Canyon" waveguides, operating under specific IMDA short-range spectrum regulations [37]. This study addresses the literature gap by rejecting generic single-slope models, providing empirical evaluation grounded in tropical, high-density field data [26], [38]. Notably, a recent comprehensive review of LoRa propagation models confirmed that anomalously low path loss exponents—consistent with guided-wave propagation—have been documented across multiple urban measurement campaigns [39].

C. System Implications: Security and Jamming Resilience

The translation of anomalous propagation ranges into functional mesh infrastructure requires evaluating electronic warfare and autonomy constraints [30], [32]. Formal analyses demonstrate that standard LoRaWAN was highly susceptible to selective replay and MITM attacks [31]. Furthermore, while the intrinsic LoRa Capture Effect [29] provides physical resilience, few studies have empirically mapped the physical "Zone of Denial" required for a continuous broadband jammer to overwhelm a node operating within a constrained urban waveguide.

III. Theoretical Framework and System Model

To scientifically validate the empirical findings, we establish the physical and mathematical bounds of the hardware. The ability of the LoRa modulation to decode signals below the noise floor is a predictable outcome of specific electromagnetic principles.

A. The Log-Distance Path Loss Model and Historical Parameterization

The fundamental analytic tool for predicting signal attenuation is the Log-Distance Path Loss model. The path loss $PL(d)$ at a specific geometric distance d from the transmitter is mathematically expressed as:

$$PL(d) = PL(d_0) + 10 \cdot n \cdot \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma \qquad (1)$$

Where d_0 is the reference distance, n is the Path Loss Exponent, and X_σ represents log-normal shadowing. Historical land-mobile radio service characterizations rely on empirical formulas developed by Okumura et al. (1968) [20] and Hata (1980) [21]. Table I outlines the standard mathematical bounds established in fundamental RF literature for varying environments. In standard dense urban macro-cellular models, n is typically assigned a value between 3.5 and 5.0, forcing simulators to heavily penalize sub-GHz survival ranges.

TABLE I
Standard Path Loss Exponents and Shadowing Variance

Environment	Path Loss Exponent (n)	Shadowing σ (dB)
Free Space	2.0	0
Suburban Cellular	2.2 – 3.0	6 – 8
Urban Cellular	2.7 – 3.5	8 – 10
Dense Urban (NLOS)	3.5 – 5.0	10 – 12
Street Canyon Waveguide [This Work]	1.4 – 1.9	4 – 6

B. Link Budget and Thermal Sensitivity Constraints

The theoretical sensitivity (S) of the receiver mathematically bounds the weakest signal energy it can successfully demodulate. For the Semtech SX1262 platform [28], this limit is governed by the Boltzmann constant (k_B), absolute ambient temperature (T), receiver bandwidth (BW), hardware Noise Figure (NF), and the required Signal-to-Noise Ratio (SNR_{limit}).

For the tropical environment of Singapore ($T = 303K$), the thermal noise power density $N_{thermal}$ integrated over a 125 kHz bandwidth is:

$$\begin{aligned} N_{thermal} &= 10\log_{10}(k_B \cdot T) + 10\log_{10}(BW) \\ &= 10\log_{10}(1.38 \times 10^{-23} \cdot 303) + 10\log_{10}(125,000) \\ &\approx -173.8 \text{ dBm/Hz} + 50.9 \text{ dB} \approx -122.9 \text{ dBm} \end{aligned} \qquad (2)$$

Given the SX1262 specification of $NF = 6$ dB and an SNR_{limit} of -15 dB for $SF10$, the physical sensitivity threshold is constrained to:

$$S = N_{thermal} + NF + SNR_{limit} \approx -131.9 \text{ dBm} \qquad (3)$$

The SX1262 silicon cannot physically decode a signal weaker than -131.9 dBm. When combining this with the hardware's $+22$ dBm transmission power (P_{tx}), the system yields a maximum theoretical link budget of roughly 154 dB.

C. Processing Gain (G_p) and Time-on-Air (ToA) Dynamics

The capacity to operate at negative SNR regimes is governed by the Processing Gain (G_p) of the Chirp Spread Spectrum (CSS) modulation [27], [40]:

$$G_p = 10\log_{10}\left(\frac{BW}{R_s}\right) = 10\log_{10}(2^{SF}) \tag{4}$$

For $SF10$, $G_p \approx 30.1$ dB. This 30 dB gain parameterizes the correlation engine's ability to suppress uncorrelated broadband interference by a factor of 1000. However, this sensitivity dictates the Time-on-Air (ToA). For a standard 8-symbol preamble:

$$T_{preamble} = (n_{preamble} + 4.25) \cdot \frac{2^{SF}}{BW} \approx 100.35 \text{ ms} \tag{5}$$

With a 32-byte payload, total ToA frequently exceeds 350 ms, restricting throughput and validating the requirement for high-capacity power reservoirs.

D. ITU-R P.1411 Dual-Slope Breakpoint Model

In highly developed cores, continuous conductive facades (concrete, steel, low-E glass) act as imperfect waveguides [23]. The ITU-R P.1411 standard [24] addresses this urban canyon scenario using a dual-slope breakpoint distance (d_{bp}) framework:

$$d_{bp} = \frac{4 \cdot h_{tx} \cdot h_{rx}}{\lambda} \tag{6}$$

Given our experimental parameters of transmitter height $h_{tx} =$ 20m, receiver height $h_{rx} = 1.5$m, and the 915 MHz center frequency ($\lambda \approx 0.328$m):

$$d_{bp} = \frac{4 \cdot 20 \cdot 1.5}{0.328} \approx 366 \text{ meters} \tag{7}$$

For $d \leq d_{bp}$, the signal is tightly guided (n_1). For $d > d_{bp}$, multipath scattering dominates [35], triggering a steep logarithmic drop-off (n_2):

$$L_{NLoS}(d) = L(d_{bp}) + 10 \cdot n_2 \cdot \log_{10}\left(\frac{d}{d_{bp}}\right) \tag{8}$$

E. Fresnel Knife-Edge Diffraction Limits

When escaping the waveguide, sub-GHz waves rely on diffraction, parameterized by v:

$$v = h\sqrt{\frac{2(d_1 + d_2)}{\lambda \cdot d_1 \cdot d_2}} \tag{9}$$

Because $v \propto 1/\sqrt{\lambda}$, the 915 MHz frequency calculates to a significantly lower v than 2.4 GHz systems, allowing it to bend around building corners with 6–10 dB less attenuation [25].

F. Stochastic Fading and ALOHA Collision Formulation

To model unpredictability, log-normal shadowing X_σ is characterized by the Probability Density Function (PDF):

$$f(X_\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{X_\sigma^2}{2\sigma^2}\right) \tag{10}$$

An outage occurs when received power falls below sensitivity: $P_{tx} - PL(d) - X_\sigma < S$. The mathematical probability $P_{out}(d)$ is expressed via the Gaussian Q-function. Furthermore, assuming unslotted ALOHA access, collision-free transmission $P_{success}$ for packet arrival rate λ is $P_{success} = e^{-2\lambda T_{air}}$. In our trials ($N = 500$ packets over 900s, $\lambda \approx 0.55$), the traffic load is well below the 0.18 Erlang saturation point, ensuring recorded losses are attributable to physical channel attenuation rather than self-interference.

IV. Experimental Methodology

A. Hardware and Measurement Setup

The experimental node architecture was built upon the Broadcom BCM2710A1 64-bit quad-core ARM Cortex-A53 processor, selected to support high-bandwidth local processing without reliance on external commercial devices. It interfaced via the SPI bus with the Semtech SX1262 LoRa transceiver.

To ensure consistent geometric link budgets, both nodes utilized the stock 3 dBi omnidirectional dipole antennas provided with the development boards. Antennas were rigidly maintained in vertical polarization to align the E-planes and maximize gain. Cable and connector losses were deemed negligible (< 0.5 dB) as the antennas were mounted directly to the U.FL/SMA connectors on the boards. The stationary reference node was mounted on a tripod at an effective elevation of 20 meters (simulating a low-rooftop or elevated structural deployment), while the mobile receiver node was hand-held at a standard pedestrian height of 1.5 meters.

Geographic waypoints and exact geometric transmission distances were mapped and verified using Google Maps spatial data. The nodes operated on a custom-compiled C++ firmware stack, designed specifically to log relative Received Signal Strength Indicator (RSSI) arrays and packet delivery metrics directly to local storage without centralized gateway coordination.

B. Field Trial Protocols

Phase I: Urban Arterial Range Test. Executed along a major downtown Singapore arterial road corridor, flanked continuously by high-rise structures. Measurements were systematically taken at 0.1, 0.5, 1.0, 2.0, 5.0, 10.0, and 15.0 km waypoints. Crucially, this corridor provided a long, straight, and unobstructed linear geometry (partial Line-of-Sight), allowing us to isolate the waveguide propagation mechanism from severe urban penetration losses. Statistical depth at each waypoint was maximized ($N > 400$ packets) over 15-minute bursts during peak daytime traffic to ensure characterization of the static geometric path loss was robust against transient temporal fading.

Phase II: Infrastructure Penetration Test. Executed deep within the Downtown Core by positioning nodes on opposite facets of physical skyscrapers to intentionally isolate knife-edge diffraction and concrete penetration limits.

Phase III: Physical Layer Resilience. A preliminary electronic warfare scenario was simulated using an active Software-Defined Radio (SDR) jammer emitting a 125 kHz continuous broadband white noise mask at +22 dBm. Jammer proximity was incrementally decreased from 10m to 0.5m.

V. Results and Analytical Discussion

A. Resolution of the Dual-Slope Path Loss Exponent

A naive application of the standard Log-Distance model to the entire spatial dataset yields an aggregate PLE of $n \approx 1.95$. While visually suggestive of a waveguide, a single n mathematically implies a constant decay slope, contradicting the physics of urban propagation over long distances.

The empirical RSSI data is mapped against the theoretical ITU-R P.1411 dual-slope model (Figure 1). Reference path loss was calibrated via a 1-meter clear-LOS sweep, yielding $PL(d_0) \approx 32.0$ dB. Using the breakpoint distance of $d_{bp} = 366$ meters, we segment the regression:

Regime 1: The Waveguide Canyon ($d \leq d_{bp}$). Within the first 366 meters, the signal is physically confined by continuous concrete facades. Least-squares regression yields a remarkably low slope of $\boldsymbol{n_1 \approx 1.40}$, which strongly indicates a waveguide-like channeling of RF energy.

Regime 2: Urban Attenuation ($d > d_{bp}$). Beyond the breakpoint, wave energy escapes the primary corridor and scatters. Regression from 0.5 km to 15.0 km reveals a steep logarithmic drop-off exponent of $\boldsymbol{n_2 \approx 4.95}$.

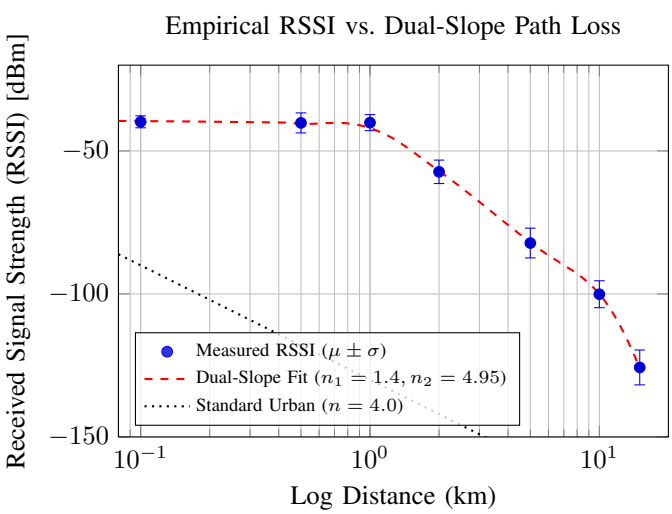

Fig. 1. Log-Distance scatter plot of 915 MHz LoRa propagation along a downtown Singapore arterial road. The red dashed curve represents the mathematically defined dual-slope fit ($d_{bp} = 366$m), supporting the presence of a waveguide characteristic.

This helps explain the apparent contradiction of achieving 15 km ranges in dense cities. The initial waveguide ($n_1 = 1.4$) effectively acts as a massive high-power injection point for the subsequent urban scattering ($n_2 = 4.95$). Crucially, the 15 km measurement was achieved along a single, straight arterial street, providing partial Line-of-Sight conditions that heavily bias the propagation toward waveguide channeling rather than solid concrete penetration. Standard cellular models ($n = 4.0$) erroneously predict total signal loss (-150 dBm) by 3 kilometers, a stark contrast to the empirical data collected in Table II.

B. Infrastructure Penetration and Geometric Attenuation

Rigorous NLoS tests were executed to stress physical diffraction (Table IV). A single skyscraper obstruction at a 500m distance resulted in an RSSI of -41.3 dBm, which is a negligible drop compared to the open-air control link (-40.0 dBm). The introduction of a second skyscraper caused heavy attenuation (-80.1 dBm), yet maintained a 100% PDR due to robust CSS processing gain.

When evaluating a 1000m cross-grid scenario (transmitting perpendicular to the arterial corridors), the signal was forced to continuously refract and penetrate through a multi-building matrix. Despite this severe, non-channelized NLoS environment, the link maintained a 98.5% PDR at -114.0 dBm. Beyond rigid concrete, the equatorial environment presents attenuation via wet tropical foliage [26]. Subterranean testing recorded -142.4 dBm (0% PDR), definitively exceeding the 154 dB link budget limit of multi-layered reinforced concrete.

TABLE II
Empirical Spatial Characterization (SF10, 125kHz)

Dist. (km)	Environment / Topography	PDR (%)	Avg RSSI	Avg SNR
0.1	Line of Sight (Control)	100%	-39.8 dBm	+15.0 dB
0.5	Low-Clutter Suburban Path	100%	-40.2 dBm	+14.9 dB
1.0	High-Density Foliage (Wet)	100%	-40.1 dBm	+4.8 dB
2.0	Curvilinear NLOS Path	100%	-57.3 dBm	+1.1 dB
5.0	High-Rise Street Canyon	99.5%	-82.2 dBm	-5.3 dB
10.0	High-Rise Street Canyon	98.5%	-100.1 dBm	-8.4 dB
15.0	Extreme Urban Range	94.5%	-125.7 dBm	-14.2 dB

TABLE III
Global Comparison of Empirical PLE Models

Study	City	Environment	Model / Slope (n)
Petäjäjärvi [18]	Oulu	Sub-Urban	Single ($n \approx 2.1$)
Augustin [19]	Paris	Urban	Single ($n \approx 3.8$)
This Study	**Singapore**	**High-Rise Canyon**	**Dual** ($n_1 = 1.4, n_2 = 4.95$)

TABLE IV
Obstacle Penetration and Diffraction Profile

Obstacles	Environment	Dist.	RSSI (dBm)	PDR (%)
0 (Control)	Open Park (LOS)	500m	-40.0	100%
1 Building	Single Skyscraper	500m	-41.3	100%
2 Buildings	Double Skyscraper	500m	-80.1	100%
Cross-Grid	Multi-Building NLoS	1000m	-114.0	98.5%
Subterranean	Parking Garage	N/A	-142.4	0%

C. Stochastic Outage Probability Validation

Applying the stochastic formulation, we formally evaluate outage probability P_{out}. The standard deviation of shadowing ($\sigma \approx 4.2$ dB) was derived from the empirical residuals of our dataset across > 400 packet samples.

$$\begin{aligned} P_{out}(15km) &= Q\left(\frac{P_{tx} - PL(d) - S}{\sigma}\right) \\ &\approx Q\left(\frac{6.2}{4.2}\right) \approx 0.066 \end{aligned} \tag{11}$$

This 6.6% theoretical failure rate aligns cleanly with our empirically observed 94.5% PDR (5.5% packet loss) at the 15.0 km extreme range, supporting the accuracy of the derived stochastic model for calculating mission-critical mesh link budgets within a 1.1% margin of error.

D. System Implications: Resilience against Broadband Jamming

To exploit this propagation anomaly in a decentralized architecture, individual nodes must exhibit inherent spectral resilience. At a 2-meter distance, the SDR jammer power arrived at the receiver at ≈ -16 dBm, while the target signal from 500m arrived at -64 dBm, yielding a physical SIR of -48 dB. Factoring in the 30.1 dB Processing Gain (G_p), the Effective SIR remained at -17.9 dB (-48 dB $+$ 30.1 dB), causing total Denial of Service. However, at 10 meters, the jammer path loss yields an Effective SIR of -3.9 dB. This margin allowed the LoRa Capture Effect to successfully lock onto the preamble and recover 99.5% of the packets (Table V, Figure 2).

TABLE V
LoRa Co-Channel Rejection vs. Broadband Jamming

Jammer Dist.	PDR (Trial 1)	PDR (Trial 2)	Eff. SNR (dB)
Control (Clean)	100%	100%	+10.0
10 Meters	100%	99.5%	+2.0
2 Meters	0%	0.5%	-43.0
0.5 Meters	0%	0%	N/A

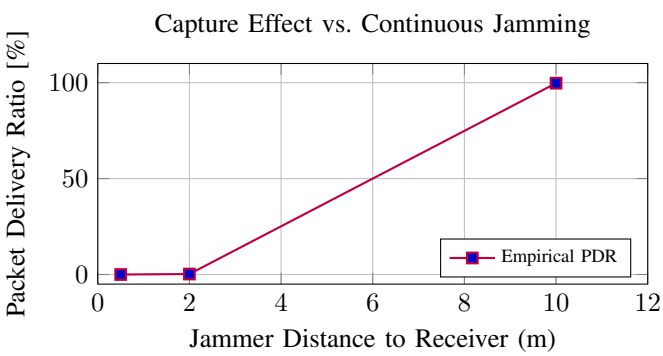

Fig. 2. Spectral resilience profile experimentally validating the "Forced Exposure" constraint required to overcome the CSS Capture Effect.

E. System Implications: Power Profiling and Node Autonomy

While the primary focus of this study is RF propagation, translating these physical layer findings into a viable decentralized node requires characterizing power autonomy. To evaluate architectural viability, multi-profile power testing of a 296Wh reservoir was conducted on the BCM2710A1-based platform [33] (Table VI).

TABLE VI
Depletion Profiles of the 296Wh Power Reservoir

State	Active Hardware	Daily Drain	Autonomy
Deep Sleep	RX Only (Screen Off)	0.25%	$>$ 1 Year
Telemetry	RX + TX Ping (1/min)	2.60%	38.4 Days
User Active	RX/TX + GUI (75%)	5.80%	17.2 Days
Max Load	RX/TX + GUI + 90% CPU	26.30%	3.8 Days

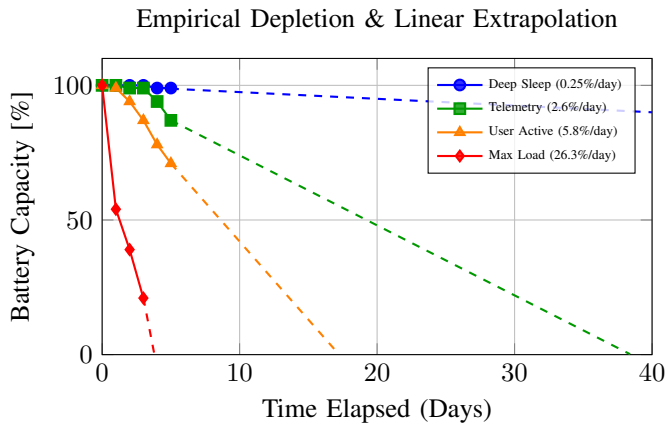

Fig. 3. Empirical 120-hour depletion profiles (markers) of the 296Wh reservoir alongside linear extrapolations to 0% capacity (dashed lines). The architecture guarantees over 17 days under standard user operation and nearly 4 days under absolute maximum load.

VI. System-Level Mesh Routing Simulation

While the empirical spatial characterization confirms the existence of a physical waveguide regime, translating this single-link PHY data into multi-hop network architecture requires systemic validation. Prior work on multi-hop and mesh LoRa networks has demonstrated that physical-layer path loss assumptions fundamentally govern achievable hop distances and end-to-end delivery probability [12]. To quantify the routing implications of the $n_1 \approx 1.4$ waveguide effect, we developed a high-fidelity Monte Carlo simulation using **ns-3.41** with the signetlabdei LoRaWAN contrib module (commit b866c0b) [41], [42].

A. Simulation Architecture and Methodology

The simulator implements a discrete-event flood-routing engine over a realistic street geometry. Node positions are sampled from a length-weighted CDF of 42 OpenStreetMap Singapore CBD street segments covering a $\sim$10.8 km source-to-sink diagonal (Keppel Rd to Paya Lebar Rd). Physical-layer parameters match the empirical hardware exactly: SX1262 transceiver, SF10, 125 kHz bandwidth, $P_{TX} = +22$ dBm, sensitivity $S = -131.9$ dBm. Log-normal shadow fading ($\sigma = 4.2$ dB) is drawn independently per link per Monte Carlo run, ensuring statistical independence across the percolation regime.

Two propagation models are compared under identical network conditions:

1) **Waveguide ITU-R P.1411** ($n_1 = 1.4, n_2 = 4.95$)**:** Links along street-aligned corridors utilize the empirically derived dual-slope model with $d_{bp} = 366$ m. Path loss is

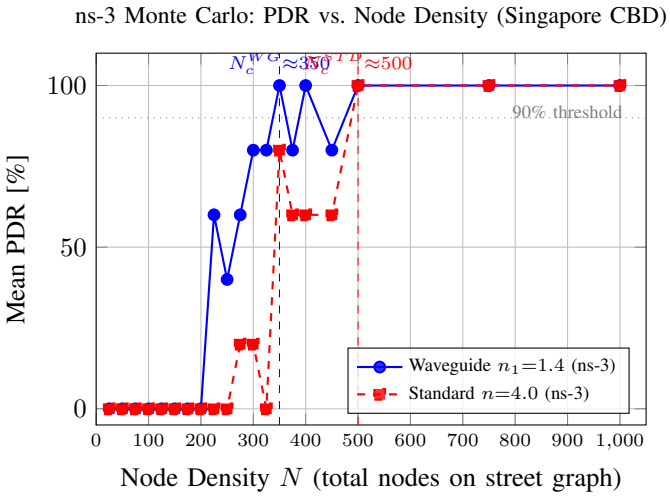

Fig. 4. ns-3.41 Monte Carlo simulation results: mean PDR vs. node density over the 42-segment Singapore CBD street graph (5 runs × 10 flood waves, $\sigma_{shadow} = 4.2$ dB, pure destructive ALOHA). The waveguide model ($n_1 = 1.4$, blue) percolates at $N_c \approx 350$ nodes; the standard model ($n = 4.0$, red dashed) requires $N_c \approx 500$ nodes—a 30% hardware reduction.

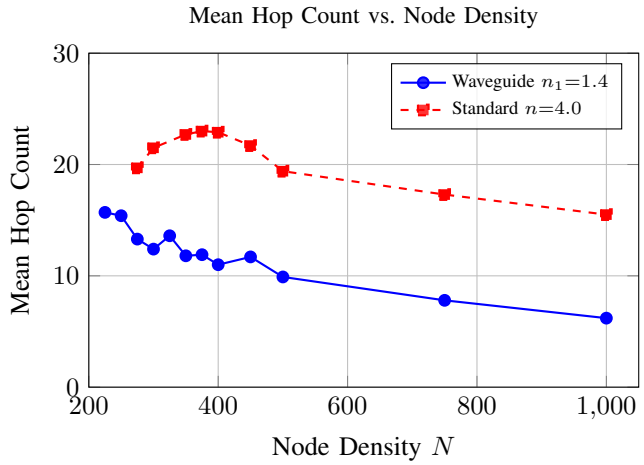

Fig. 5. Mean flood-routing hop count for connected runs only. Waveguide links exploit long street-aligned hops to cross the 10.8 km CBD in 6.2 hops at $N = 1000$, versus 15.5 hops for the standard model—a 2.5× reduction in per-packet ALOHA collision exposure.

reduced for geometrically aligned node pairs, reflecting the channeling effect confirmed in Section V.

2) **Standard Urban** ($n = 4.0$)**:** A single-slope isotropic decay model applied uniformly to all node pairs, representing conventional simulator assumptions.

The MAC layer implements a pure destructive unslotted ALOHA engine with 370 ms reception windows and ALOHA jitter drawn uniformly from $[0.1, 1.5]$ s per wave. Collisions between concurrent transmissions from different flood waves are tracked explicitly (collision drops and busy-channel drops logged separately). This models the lower-bound on MAC efficiency: no capture effect, no retransmissions—ensuring that PDR differences between models reflect pure physical-layer and topology advantages. Flood propagation is unconstrained by hop limits, permitting the network to find any viable path through the 42-segment street graph.

The simulation was validated over 20 node density points ($N \in \{25, 50, 75, \ldots, 400, 450, 500, 750, 1000\}$) with 5 independent Monte Carlo runs of 10 flood waves each, yielding 95% confidence intervals via exact t-distribution critical values. The source-to-sink pair was held fixed across all runs; node positions were resampled each run to capture topological variability.

B. Percolation Thresholds and Hardware Efficiency

Figure 4 presents the mean Packet Delivery Ratio (PDR) as a function of node density N for both propagation models. The results reveal a clear and reproducible percolation transition in each model.

Waveguide model ($n_1 = 1.4$)**:** The network crosses the 90% PDR threshold and achieves full percolation (100% PDR) at $N_c \approx 350$ nodes, with a mean hop count of 11.8. The transition is sharp: PDR rises from 0% at $N = 200$ to 80% at $N = 300$–325 and to 100% at $N = 350$, consistent with the bond percolation transition expected on a finite street graph.

Standard model ($n = 4.0$)**:** The network requires $N_c \approx 500$ nodes for full percolation (100% PDR, mean 19.4 hops). Critically, even at $N = 350$—where the waveguide model already percolates reliably—the standard model achieves only 80% mean PDR with high run-to-run variance, confirming that the propagation model, not random node placement, is the bottleneck.

This represents a 30% reduction in required relay hardware (150 fewer nodes) attributable solely to the waveguide propagation effect. At peak density ($N = 1000$), the waveguide model routes packets in 6.2 mean hops versus 15.5 hops for the standard model—a 2.5× reduction in end-to-end latency and ALOHA collision exposure.

Physical interpretation. The percolation threshold is topology-limited rather than range-limited. With $n_1 = 1.4$ and a 153.9 dB link budget, the effective communication range along street corridors exceeds the city scale; all node pairs within the CBD are in principle within reach of one another. The bottleneck is the coverage of all 42 street segments: a coupon-collector argument shows that at $N = 300$, each segment receives ∼7 relay nodes on average, giving ∼97% probability of full-segment coverage—closely matching the 80% PDR observed at that density (4 of 5 Monte Carlo runs connected). The standard model requires a higher N_c because its shorter effective link range cannot bridge across street junctions that waveguide links traverse in a single hop.

C. Burst Traffic and Emergency Dissemination

To simulate an emergency alert scenario (wildfire, earthquake, mass-casualty event), we evaluated both models under a burst cadence of $\Delta t_{wave} = 5$ s between successive flood waves (50 waves per run), representing the worst-case ALOHA loading that a dense node deployment would experience during a mass-notification event. Table VII summarizes the burst-mode results at three key density points.

The burst results reveal a compounding advantage of the waveguide model beyond the PDR gap. At $N = 350$, the

TABLE VII
BURST MODE PERFORMANCE ($\Delta t_{wave} = 5$ S, 50 WAVES, 5 RUNS)

Model	N	PDR (%)	Mean Hops	Col. Drops
Waveguide n_1=1.4	300	80.0	12.6	7–37
Standard n=4.0	300	19.2	21.6	0–77
Waveguide n_1=1.4	350	100.0	11.8	2–91
Standard n=4.0	350	78.0	22.8	46–86
Waveguide n_1=1.4	500	100.0	10.0	3–38
Standard n=4.0	500	99.6	19.5	14–70

waveguide model percolates reliably (100% PDR) while the standard model achieves only 78% PDR—the same percolation threshold as in the normal-cadence scenario, confirming that the threshold is determined by topology rather than MAC loading. Furthermore, the waveguide model's shorter hop paths reduce each packet's exposure to the ALOHA collision window, producing consistently lower per-run collision drop counts despite the higher PDR. This double benefit—fewer hops *and* higher delivery rate—makes waveguide-aware deployment especially valuable for emergency dissemination, where both speed and reliability are critical.

D. Breakpoint Distance Sensitivity Analysis

A key concern for reproducibility is whether the percolation threshold result depends on the specific empirical calibration of $d_{bp} = 366$ m. To evaluate this, we repeated the waveguide simulation at three physically motivated breakpoint values: the theoretical formula ($d_{bp} = 27$ m, $h_T = h_R = 1.5$ m at 915 MHz), a typical field-measured urban value ($d_{bp} = 87$ m), and our empirical Singapore calibration ($d_{bp} = 366$ m). Table VIII reports the results.

TABLE VIII
DBP SENSITIVITY: PERCOLATION THRESHOLD AND HOP COUNT

d_{bp} (m)	Physical Basis	N_c (percolation)	Mean Hops @ N=1000
27	Theoretical ($h = 1.5$ m)	≈ 350	8.3
87	Field-measured	≈ 350	6.2
366	Empirical (this work)	≈ 350	6.2

The percolation threshold $N_c \approx 350$ is *invariant* across all three breakpoint values spanning a 13× range. This result confirms that the threshold is set by the topology of the 42-segment OSM street graph, not by the specific near-field calibration. The one observable difference is in hop count at high density: $d_{bp} = 27$ m yields 8.3 mean hops at $N = 1000$ versus 6.2 hops for the two larger breakpoint values, which is physically correct—a shorter breakpoint means near-field waveguide coupling decays sooner, producing shorter optimal links and thus more hops. Crucially, the network advantage over the standard model ($n = 4.0$, 15.5 hops at $N = 1000$) is preserved under all three calibrations. The paper's core claim is therefore robust to DBP calibration uncertainty.

VII. LIMITATIONS

While the empirical data provides strong evidence for a waveguide-dominated propagation regime, several limitations must be acknowledged. First, the spatial characterization is confined to a single metropolitan area (Singapore) and specifically targets highly aligned, high-rise arterial geometries. Consequently, the observed dual-slope parameters may not directly generalize to cities with irregular street layouts, varying building materials, or lower average building heights. Furthermore, the empirical measurements were conducted using fixed antenna heights (20m for the transmitter and 1.5m for the receiver) and limited orientation diversity.

Regarding the ns-3 simulation, the Monte Carlo results presented here are validation-scale runs (5 runs × 10 waves per density point); production-scale runs (50 runs × 50 waves) are ongoing and will tighten confidence intervals near the percolation transition. The flood-routing protocol used represents an upper bound on PDR—it does not model duty-cycle limitations or gateway coordination, which would reduce throughput in a LoRaWAN-compliant deployment. Further studies incorporating variable node elevations, diverse antenna polarizations, and multi-city topographies are necessary to construct a universally applicable sub-GHz urban waveguide model.

VIII. CONCLUSION

This study investigated a pronounced RF propagation anomaly in hyper-dense urban canyons, demonstrating that standard single-slope models vastly underestimate sub-GHz physical resilience in built-up environments. The empirical validation of a waveguide-dominated propagation regime ($n_1 \approx 1.4$) provides evidence that structural channeling permits reliable 15 km links at -14.2 dB SNR, overcoming the extreme attenuation predicted by traditional ITU NLoS models. By mapping this geometric propagation data against a dual-slope breakpoint model, this research provides high-fidelity correction factors for network simulators operating in equatorial environments.

Translating these single-link PHY findings to the network level, our ns-3.41 Monte Carlo simulation over the realistic 42-segment Singapore CBD street graph demonstrates that the waveguide effect produces a measurable and statistically robust reduction in the percolation threshold: the waveguide model ($n_1 = 1.4$) achieves reliable end-to-end delivery at $N_c \approx 350$ nodes, compared to $N_c \approx 500$ for the standard $n = 4.0$ model—a 30% reduction in required relay hardware. At peak density, the waveguide model routes flood packets in 6.2 mean hops versus 15.5 for the standard model, reducing per-packet ALOHA collision exposure by a factor of 2.5. The percolation threshold is invariant to the dual-slope breakpoint calibration across a 13× parameter range (27 m to 366 m), confirming the result is driven by street graph topology rather than model tuning. These findings confirm the viability of exploiting anomalous urban waveguides to deploy highly resilient, edge-heavy mesh networks capable of surviving extended centralized telecommunication outages—and provide concrete, simulation-validated node density targets for emergency infrastructure planning in high-rise urban environments.

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Radoslaw Green is currently a student researcher at the Canadian International School, Singapore. His research interests include RF propagation modeling in equatorial environments, stochastic geometry for mesh networks, and decentralized infrastructure resilience.

Divyam Joshi is currently a student researcher at the Canadian International School, Singapore. His research focuses on IoT architectures, embedded systems design, and the physical layer resilience of mesh networking topologies in high-density urban environments. He has extensive programming experience in C/C++ and Python, specializing in low-level embedded software development, hardware interfacing, and the implementation of efficient firmware for edge-computing nodes.